

# SMART PARKING SYSTEM

## System Modeling & Design Review Report

Requirements • Use Cases • Activity • Sequences • Architecture • State Machines

| Field          | Value                                  |
|----------------|----------------------------------------|
| Project        | Smart Parking System                   |
| Modeling Tool  | TTool / AVATAR                         |
| Source Model   | Smart_Parking_TTool.xml                |
| Prepared for   | Embedded Systems Project Documentation |
| Author         | Mohamed Aldreamly                      |
| Report Version | 1.0                                    |
| Date           | 29 September 2026                      |

### Scope statement

This report is based on the supplied TTool XML model and documents what is represented in that model. It does not mark requirements as verified or satisfied beyond the status recorded in the source model.

## 1. Executive Summary

The TTool model defines a three-ECU Smart Parking System built around one STM32 central ECU and two ATmega32 gate ECUs for Entry and Exit. The model covers requirements, behavioral analysis, interaction sequences, architectural decomposition, and gate-control state machines.

The requirements view contains 49 requirement nodes in total: one system-level root, eight requirement-category containers, and 40 detailed requirements. All requirement nodes are currently recorded in the XML with satisfied=false and verified=false; therefore the model should be treated as specified but not formally verified in TTool.

The analysis view contains one system Use Case Diagram, one Entry Vehicle Activity Diagram, and three Sequence Diagrams covering Entry, Exit, and Authentication. The design view contains the central STM32 ECU, Entry and Exit ATmega32 ECUs, and the internal functional blocks EntryControl, ExitControl, Communication\_Manager, AccessControl, and ParkingDataManager.

The strongest part of the model is the consistency of the core access-control concept: User ID is checked before RFID, RFID mismatch is retried up to three times, the STM32 makes the final access decision, parking capacity and user INSIDE/OUTSIDE state are considered, and the gate is kept open while a vehicle is detected.

## 2. Model Inventory

| Model Element        | Count | Coverage                                                                                |
|----------------------|-------|-----------------------------------------------------------------------------------------|
| Requirements         | 1     | AVATARRD requirement model                                                              |
| Use Case Diagrams    | 1     | Smart Parking System - Use Case Diagram                                                 |
| Activity Diagrams    | 1     | Entry Vehicle Activity Diagram                                                          |
| Sequence Diagrams    | 3     | Entry Vehicle, Exit Vehicle, Authentication                                             |
| Block Diagrams       | 1     | System architectural decomposition                                                      |
| State-Machine Panels | 8     | EntryControl and ExitControl populated; other panels present as structural placeholders |

## 3. Requirements Model Review

The requirement model is organized hierarchically under Smart\_Parking\_System. The detailed requirements are grouped into Authentication, User Registration, Parking Management, Gate Control, Persistence, Communication, User Interface, and System Architecture.

| Group              | Detailed Requirements | Primary Concern                                    |
|--------------------|-----------------------|----------------------------------------------------|
| Authentication     | 6                     | ID/RFID authentication and retry handling          |
| User Registration  | 4                     | Authorized registration and persistent credentials |
| Parking Management | 7                     | Capacity, occupancy, and user parking state        |
| Gate Control       | 6                     | Entry/Exit gate control and safe                   |

|                     |   |                                                  |
|---------------------|---|--------------------------------------------------|
|                     |   | closing                                          |
| Persistence         | 4 | EEPROM persistence and startup recovery          |
| Communication       | 5 | UART communication and central decisions         |
| User Interface      | 4 | Keypad, RFID, feedback, and parking status       |
| System Architecture | 4 | STM32/ATmega32 ECU allocation and centralization |

### 3.1 Detailed Requirement Register

| ID          | Requirement             | Model Description                                                                                                                                              | Satisfied / Verified |
|-------------|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| REQ-AUTH-01 | User_Identification     | The system shall accept a User ID before performing RFID authentication.                                                                                       | false/false          |
| REQ-AUTH-02 | RFID_Authentication     | The system shall authenticate a user using the RFID card associated with the entered User ID.                                                                  | false/false          |
| REQ-AUTH-03 | ID_Validation           | The system shall reject an unregistered User ID.                                                                                                               | false/false          |
| REQ-AUTH-04 | Card_Mismatch           | If the User ID exists but the scanned RFID card does not belong to that user, the system shall reject the card and inform the user that the card is incorrect. | false/false          |
| REQ-AUTH-05 | Retry_Limit             | The system shall allow a maximum of three incorrect RFID attempts for a valid User ID during one authentication session.                                       | false/false          |
| REQ-AUTH-06 | Access_Denial           | After three unsuccessful RFID attempts, the system shall deny access and terminate the authentication session.                                                 | false/false          |
| REQ-REG-01  | User_Registration       | The system shall allow an authorized operator to register new users.                                                                                           | false/false          |
| REQ-REG-02  | User_Credentials        | Each registered user shall have a unique User ID associated with an RFID card UID.                                                                             | false/false          |
| REQ-REG-03  | Multiple_Users          | The number of registered users may exceed the physical parking capacity.                                                                                       | false/false          |
| REQ-REG-04  | Persistent_Registration | Registered user information shall remain available after power loss or system restart.                                                                         | false/false          |
| REQ-PARK-01 | Parking_Capacity        | The initial parking capacity shall be three vehicles.                                                                                                          | false/false          |
| REQ-PARK-02 | Logical_Occupancy       | Parking occupancy shall be determined logically from successful entry and exit operations without individual slot sensors.                                     | false/false          |
| REQ-PARK-03 | User_Parking_State      | The system shall maintain whether each registered user is INSIDE or OUTSIDE.                                                                                   | false/false          |
| REQ-PARK-04 | Full_Parking            | The system shall deny vehicle entry when the parking is full even when the user's credentials are valid.                                                       | false/false          |
| REQ-PARK-05 | Duplicate_Entry         | The system shall prevent a user whose state is INSIDE from entering again.                                                                                     | false/false          |
| REQ-PARK-06 | Invalid_Exit            | The system shall prevent an OUTSIDE user from performing a valid exit operation.                                                                               | false/false          |

|             |                              |                                                                                                                 |             |
|-------------|------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------|
| REQ-PARK-07 | Occupancy_Update             | The system shall update parking occupancy only after a successful vehicle entry or exit.                        | false/false |
| REQ-GATE-01 | Entry_Gate                   | The Entry ECU shall control the entry gate.                                                                     | false/false |
| REQ-GATE-02 | Exit_Gate                    | The Exit ECU shall control the exit gate.                                                                       | false/false |
| REQ-GATE-03 | Authorized_Opening           | A gate shall open only after receiving a valid access decision from the central controller.                     | false/false |
| REQ-GATE-04 | Vehicle_Detection            | Each gate shall contain a digital vehicle-presence or obstruction sensor.                                       | false/false |
| REQ-GATE-05 | Safe_Closing                 | The system shall not close a gate while a vehicle is detected beneath the gate.                                 | false/false |
| REQ-GATE-06 | Automatic_Closing            | The gate shall close automatically after the vehicle has passed and the gate area becomes clear.                | false/false |
| REQ-MEM-01  | Non_Volatile_Storage         | The STM32 Admin ECU shall use EEPROM for persistent system data.                                                | false/false |
| REQ-MEM-02  | User_Database                | The system shall persist User ID and associated RFID UID information.                                           | false/false |
| REQ-MEM-03  | Parking_State                | The system shall preserve the required parking and user state to recover operation after a power cycle.         | false/false |
| REQ-MEM-04  | Startup_Recovery             | After startup, the STM32 shall restore the required persistent state before allowing normal parking operations. | false/false |
| REQ-COM-01  | ECU_Communication            | The STM32 Admin ECU shall communicate with the Entry and Exit ATmega32 ECUs using UART.                         | false/false |
| REQ-COM-02  | Entry_Authentication_Request | The Entry ECU shall send the entered User ID and scanned RFID information to the STM32 for authentication.      | false/false |
| REQ-COM-03  | Exit_Authentication_Request  | The Exit ECU shall send the entered User ID and scanned RFID information to the STM32 for authentication.       | false/false |
| REQ-COM-04  | Central_Decision             | The STM32 shall make the final access decision.                                                                 | false/false |
| REQ-COM-05  | Authentication_Response      | The STM32 shall return an authentication result to the requesting gate ECU.                                     | false/false |
| REQ-UI-01   | User_ID_Input                | The system shall provide a keypad interface for entering a User ID.                                             | false/false |
| REQ-UI-02   | RFID_Input                   | The system shall provide an RFID interface for scanning user cards.                                             | false/false |
| REQ-UI-03   | User_Feedback                | The system shall display feedback corresponding to the result of an access request.                             | false/false |
| REQ-UI-04   | Parking_Status               | The system shall provide information about current parking availability.                                        | false/false |
| REQ-SYS-01  | Central_ECU                  | The system shall use an STM32-based ECU as the central controller.                                              | false/false |
| REQ-SYS-02  | Entry_ECU                    | The system shall use an ATmega32-based ECU for entry-gate control.                                              | false/false |
| REQ-SYS-03  | Exit_ECU                     | The system shall use an ATmega32-based ECU for exit-gate control.                                               | false/false |
| REQ-SYS-04  | Centralized_Authentication   | User database management, parking-state management, and final authentication decisions                          | false/false |

|  |  |                                            |  |
|--|--|--------------------------------------------|--|
|  |  | shall be handled by the STM32 central ECU. |  |
|--|--|--------------------------------------------|--|

### 3.2 Requirements Observations

- All detailed requirements are currently marked satisfied=false and verified=false in the source model.
- REQ-AUTH-03 (ID\_Validation) contains an incomplete description ending after “The system shall reject an unregistered”.
- REQ-COM-03 (Exit\_Authentication\_Request) contains an incomplete description ending after “The Exit ECU shall send the entered User”.
- REQ-SYS-03 (Exit\_ECU) contains only “for exit-gate control.” and is missing the opening part of the sentence.
- The category node Authentication\_Requirements still contains the default placeholder text “Requirement description: Double-click to edit”.
- The detailed requirements nevertheless establish the main functional baseline: centralized authentication, UART ECU communication, persistent storage, logical occupancy, and gate safety.

## 4. Use Case Model Review

The Use Case Diagram defines the system boundary as Smart Parking System and identifies two external actors.

| Actors                   | Use Cases                           |
|--------------------------|-------------------------------------|
| Driver / Authorized User | Primary parking user                |
| Administrator            | Administrative / authorization role |

### 4.1 Modeled Use Cases

- Authenticate ID + RFID
- Enter Parking
- Exit Parking
- Check Parking Availability
- Validate Entry Eligibility
- Validate Exit Eligibility
- Operate Gate Safely
- View Access / Parking Status
- Register Authorized User
- Manage User Authorization
- Track Parking Occupancy
- Preserve / Restore System State

The use-case set covers the complete functional intent represented elsewhere in the model: entry, exit, authentication, eligibility checks, gate safety, registration/authorization, occupancy tracking, status feedback, and persistence/recovery. The XML also records <<include>> relationships between use cases.

## 5. Entry Vehicle Activity Diagram Review

The Entry Vehicle Activity Diagram models the operational flow from User ID entry through authentication, parking eligibility, gate operation, vehicle detection, state update, and timeout handling.

1. Enter User ID through keypad and validate that the ID is registered.
2. Request and read RFID, then match the RFID UID with the User ID.
3. On mismatch, increment RFID attempts and allow retries while attempts are below three.
4. Deny access after the retry limit is reached.
5. Check parking-full condition before granting entry.
6. On access grant, open the Entry gate and start a 10-second entry timer.
7. Detect the vehicle using the IR sensor; keep the gate open while the sensor is BLOCKED.
8. Wait until the gate sensor is CLEAR before closing.
9. After successful passage, set the user state to INSIDE, increment occupied count, save required state to EEPROM, and display updated parking status.
10. If no vehicle is detected within 10 seconds, close the gate and display Entry Timeout.

This activity model explicitly captures the safety condition that the gate remains open while the vehicle sensor is blocked, and it separates an authenticated access decision from a completed physical vehicle passage.

## 6. Sequence Diagram Review

### 6.1 Entry Vehicle Sequence Diagram

The Entry sequence starts with enterUserID(userID), forwards AUTH\_ID\_REQUEST(userID), reads the user record, handles invalid ID, requests RFID for a valid ID, checks RFID ownership and retry limits, evaluates INSIDE state and occupiedCount, grants or denies entry, operates the gate, and finally sends ENTRY\_COMPLETED(userID) before persisting the INSIDE state.

| #  | Modeled Interaction                  |
|----|--------------------------------------|
| 1  | enterUserID(userID)                  |
| 2  | AUTH_ID_REQUEST(userID)              |
| 3  | findUser(userID)                     |
| 4  | userRecord / NOT_FOUND               |
| 5  | [invalid ID] INVALID_ID              |
| 6  | [invalid ID] display Invalid ID      |
| 7  | [valid ID] REQUEST_RFID              |
| 8  | presentRFID(uid)                     |
| 9  | RFID_UID(uid)                        |
| 10 | matchRFID(userID, uid)               |
| 11 | matchResult                          |
| 12 | [wrong RFID && attempts<3] TRY_AGAIN |
| 13 | display Wrong Card - Try Again       |
| 14 | [attempts==3] ACCESS_DENIED          |
| 15 | readUserStateAndOccupancy()          |
| 16 | state, occupiedCount                 |
| 17 | [already INSIDE] ALREADY_INSIDE      |
| 18 | [occupiedCount>=3] PARKING_FULL      |
| 19 | [eligible] ACCESS_GRANTED            |
| 20 | display Access Granted               |
| 21 | OPEN_GATE                            |

|    |                                   |
|----|-----------------------------------|
| 22 | KEEP_OPEN                         |
| 23 | CLOSE_GATE                        |
| 24 | ENTRY_COMPLETED(userID)           |
| 25 | set INSIDE; occupiedCount++; save |

## 6.2 Exit Vehicle Sequence Diagram

The Exit sequence mirrors authentication but evaluates the exit rule instead of parking capacity. An OUTSIDE user receives INVALID\_EXIT; an INSIDE user can receive ACCESS\_GRANTED. After gate passage, EXIT\_COMPLETED(userID) is sent and the persistent state is updated to OUTSIDE.

| #  | Modeled Interaction                     |
|----|-----------------------------------------|
| 1  | enterUserID(userID)                     |
| 2  | AUTH_ID_REQUEST(userID)                 |
| 3  | findUser(userID)                        |
| 4  | userRecord / NOT_FOUND                  |
| 5  | [invalid ID] INVALID_ID                 |
| 6  | [invalid ID] display Invalid ID         |
| 7  | [valid ID] REQUEST_RFID                 |
| 8  | presentRFID(uid)                        |
| 9  | RFID_UID(uid)                           |
| 10 | matchRFID(userID, uid)                  |
| 11 | matchResult                             |
| 12 | [wrong RFID && attempts<3] TRY_AGAIN    |
| 13 | display Wrong Card - Try Again          |
| 14 | [attempts==3] ACCESS_DENIED             |
| 15 | readUserState()                         |
| 16 | userState                               |
| 17 | [user OUTSIDE] INVALID_EXIT             |
| 18 | [user INSIDE] ACCESS_GRANTED            |
| 19 | display Access Granted                  |
| 20 | OPEN_GATE                               |
| 21 | BLOCKED                                 |
| 22 | CLEAR                                   |
| 23 | EXIT_COMPLETED(userID)                  |
| 24 | set OUTSIDE; occupiedCount--; save      |
| 25 | EXIT_CONFIRMED / display Exit Completed |

## 6.3 Authentication Sequence Diagram

The Authentication sequence isolates the reusable authentication behavior. It checks the User ID, requests RFID, increments the attempt counter on mismatch, permits retry while attempts are below three, denies after the third failed attempt, and evaluates Entry/Exit eligibility after a successful UID match.

| #  | Modeled Interaction               |
|----|-----------------------------------|
| 1  | enterUserID(userID)               |
| 2  | AUTH_ID_REQUEST(userID)           |
| 3  | findUser(userID)                  |
| 4  | userRecord / NOT_FOUND            |
| 5  | [ID not registered] INVALID_ID    |
| 6  | display Invalid ID                |
| 7  | [ID valid] REQUEST_RFID           |
| 8  | presentRFID(uid)                  |
| 9  | AUTH_RFID_REQUEST(userID, uid)    |
| 10 | getExpectedRFID(userID)           |
| 11 | expectedUID                       |
| 12 | [UID mismatch] incrementAttempt() |

|    |                                            |
|----|--------------------------------------------|
| 13 | [attempts < 3] WRONG_CARD / RETRY          |
| 14 | display Wrong Card - Try Again             |
| 15 | [retry] presentRFID(newUID)                |
| 16 | AUTH_RFID_REQUEST(userID, newUID)          |
| 17 | [attempts == 3] ACCESS_DENIED              |
| 18 | display Access Denied                      |
| 19 | [UID matches] resetAttemptCounter()        |
| 20 | readUserStateAndOccupancy()                |
| 21 | state, occupiedCount                       |
| 22 | [Entry && INSIDE] ALREADY_INSIDE           |
| 23 | [Entry && occupiedCount >= 3] PARKING_FULL |
| 24 | [Exit && OUTSIDE] INVALID_EXIT             |
| 25 | [eligible] AUTH_SUCCESS                    |
| 26 | return authentication result               |
| 27 | display final access status                |

## 6.4 Sequence-Model Observation

The XML stores some lifeline captions through TTool-specific infoparam fields whose field names do not always correspond cleanly to the visible semantic role. The interaction messages themselves are clear and were used as the authoritative basis for this review. A final visual pass inside TTool is recommended before submission to confirm that each visible lifeline caption is positioned and named as intended.

## 7. System Architecture / Block Diagram Review

The AVATAR Block Diagram represents a distributed three-ECU architecture. The Entry and Exit Atmega32 ECUs handle gate-side interaction, while the STM32 Central ECU contains the centralized parking and authentication logic.

| Block                 | Role                     | Responsibility Represented by Model                                      |
|-----------------------|--------------------------|--------------------------------------------------------------------------|
| STM32_Central_ECU     | Central controller       | Contains Communication_Manager, AccessControl, and ParkingDataManager.   |
| Entry_Atmega32_ECU    | Entry gate ECU           | Contains EntryControl and performs entry-side interaction/gate behavior. |
| Exit_Atmega32_ECU     | Exit gate ECU            | Contains ExitControl and performs exit-side interaction/gate behavior.   |
| Communication_Manager | Central functional block | Routes authentication/completion requests and central responses.         |
| AccessControl         | Central functional block | Represents centralized authentication and eligibility decisions.         |
| ParkingDataManager    | Central functional block | Represents parking/user data management and persistent-state access.     |
| EntryControl          | Entry functional block   | Implements the Entry control state machine.                              |
| ExitControl           | Exit functional block    | Implements the Exit control state machine.                               |

The block diagram includes UART Connection notes between the distributed ECUs, consistent with the communication requirements that specify UART between the STM32 Admin/Central ECU and both Atmega32 gate ECUs.

## 8. State Machine Review

### 8.1 EntryControl

| Modeled States            |
|---------------------------|
| Entry_StateCloseGate      |
| Entry_StateVehiclePassing |
| Entry_StateWaitVehicle    |
| Entry_StateGateOpen       |
| Entry_StateWaitRFIDResult |
| Entry_StateReadRFID       |
| Entry_StateWaitIDResult   |
| Entry_StateReadID         |
| Entry_StateIdle           |

EntryControl progresses from Idle to ID acquisition/validation, RFID acquisition/validation, gate opening, vehicle waiting/passage, and gate closing. The model also includes authentication request signals, access-result signals, and a 10-second timer.

| Signals / Timer Operations |
|----------------------------|
| expire(timer1)             |
| reset(timer1)              |
| set(timer1, 10)            |
| parkingFull()              |
| accessDenied()             |
| authRFIDReq(id, uid)       |
| accessGranted()            |
| accessDenied()             |
| wrongCard()                |
| idValid()                  |
| invalidID()                |
| authIDReq(userID)          |

### 8.2 ExitControl

| Modeled States           |
|--------------------------|
| Exit_StateCloseGate      |
| Exit_StateVehiclePassing |
| Exit_StateWaitVehicle    |
| Exit_StateGateOpen       |
| Exit_StateWaitRFIDResult |
| Exit_StateReadRFID       |
| Exit_StateWaitIDResult   |
| Exit_StateReadID         |
| Exit_StateIdle           |

ExitControl follows the same structural pattern as EntryControl, with the exit-specific invalidExit() response and the same 10-second timer mechanism for gate-side passage handling.

| Signals / Timer Operations |
|----------------------------|
| expire(timer1)             |
| reset(timer1)              |

|                      |
|----------------------|
| set(timer1, 10)      |
| invalidExit()        |
| accessDenied()       |
| authRFIDReq(id, uid) |
| accessGranted()      |
| accessDenied()       |
| wrongCard()          |
| idValid()            |
| invalidID()          |
| authIDReq(userID)    |

### 8.3 Structural State-Machine Panels

The model contains state-machine panels for STM32\_Central\_ECU, Entry\_ATmega32\_ECU, Exit\_ATmega32\_ECU, Communication\_Manager, AccessControl, ParkingDataManager. These panels do not contain behavioral states/signals in the supplied XML. This is consistent with treating several of these elements as architectural/service blocks rather than gate-session FSMs, but the empty panels should be recognized as structural placeholders rather than completed behavioral state machines.

## 9. Cross-Model Traceability

| Concern                      | Requirements           | Use Case                        | Behavioral Evidence                        | Design Allocation                       |
|------------------------------|------------------------|---------------------------------|--------------------------------------------|-----------------------------------------|
| ID before RFID               | REQ-AUTH-01/02         | Authenticate ID + RFID          | Entry Activity;<br>Authentication Sequence | EntryControl/ExitControl                |
| Max. 3 wrong RFID attempts   | REQ-AUTH-05/06         | Authenticate ID + RFID          | Entry Activity;<br>Authentication Sequence | wrongCard()/<br>accessDenied() handling |
| Parking capacity = 3         | REQ-PARK-01/04         | Check Parking Availability      | Entry Sequence                             | Central eligibility decision            |
| INSIDE / OUTSIDE state       | REQ-PARK-03/05/06      | Validate Entry/Exit Eligibility | Entry/Exit Sequences                       | AccessControl +<br>ParkingDataManager   |
| Safe gate closing            | REQ-GATE-04/05/06      | Operate Gate Safely             | Entry Activity; Entry/Exit Sequences       | EntryControl/ExitControl                |
| Persistent user/system state | REQ-MEM-01..04         | Preserve / Restore System State | Entry/Exit Sequences                       | ParkingDataManager                      |
| UART ECU communication       | REQ-COM-01..05         | System interactions             | All sequences                              | Communication_Manager                   |
| Centralized decision         | REQ-COM-04; REQ-SYS-04 | Eligibility use cases           | Authentication/Entry/Exit sequences        | STM32 Central +<br>AccessControl        |

## 10. Professional Review Findings

### 10.1 Strengths

- Clear separation between central decision-making and gate-side control.
- Authentication is modeled as ID-first followed by RFID verification.
- Retry-limit behavior is explicitly represented rather than left implicit.
- Entry eligibility distinguishes valid credentials from parking-full and already-inside conditions.
- Exit eligibility explicitly rejects an OUTSIDE user through INVALID\_EXIT.
- Gate safety is represented through vehicle detection, BLOCKED/CLEAR behavior, and delayed closing.
- Persistent state is part of both requirements and interaction flows.
- The architecture names the major functional blocks used by the implementation-oriented design.

## 10.2 Items to Correct or Complete Before Final Submission

- Complete the truncated requirement descriptions in REQ-AUTH-03, REQ-COM-03, and REQ-SYS-03.
- Replace the default placeholder text under Authentication\_Requirements.
- When verification/testing is complete, update the requirement satisfied/verified status in TTool; the current XML records every requirement as false/false.
- Visually confirm the lifeline captions in the three Sequence Diagrams because the XML contains TTool-specific caption-field mappings that are semantically awkward when read as raw XML.
- Keep the empty state-machine panels only if they are intentionally structural/service placeholders; otherwise populate them before claiming complete behavioral modeling for those blocks.
- Ensure the final TTool model and the final implementation remain synchronized if any protocol, timing, registration, or persistence behavior changes.

## 11. Final Assessment

The supplied TTool model provides a coherent system-level design for the Smart Parking System. It establishes the functional requirements, defines the external actors and principal use cases, models the main Entry behavior, documents Entry/Exit/Authentication interactions, decomposes the software across the three ECUs, and provides explicit EntryControl and ExitControl state machines.

From a documentation perspective, the model is suitable as a strong design baseline. The remaining work is primarily model finalization rather than architectural redesign: complete the few truncated requirement texts, remove placeholder requirement text, confirm sequence lifeline labels visually in TTool, and update requirement verification/satisfaction status after the corresponding implementation and tests have been completed.

Accordingly, this report should be read as a formal review of the current model content, not as evidence that the implementation has passed verification or that every requirement has been validated.

## Appendix A - Diagram Register

| Type          | Diagram / Panel                         | Purpose / Status                           |
|---------------|-----------------------------------------|--------------------------------------------|
| Requirements  | AVATARRD                                | System and detailed requirements           |
| Use Case      | Smart Parking System - Use Case Diagram | Actors and system functions                |
| Activity      | Entry Vehicle Activity Diagram          | Entry operational flow and gate safety     |
| Sequence      | Entry Vehicle Sequence Diagram          | End-to-end Entry interaction               |
| Sequence      | Exit Vehicle Sequence Diagram           | End-to-end Exit interaction                |
| Sequence      | Authentication Sequence Diagram         | Reusable authentication behavior           |
| Design        | Block Diagram                           | ECU and functional-block architecture      |
| State Machine | STM32_Central_ECU                       | Panel present; no behavioral states in XML |
| State Machine | Entry_ATmega32_ECU                      | Panel present; no behavioral states in XML |
| State Machine | Exit_ATmega32_ECU                       | Panel present; no behavioral states in XML |
| State Machine | EntryControl                            | Behavioral FSM                             |
| State Machine | Communication_Manager                   | Panel present; no behavioral states in XML |
| State Machine | AccessControl                           | Panel present; no behavioral states in XML |

|               |                    |                                            |
|---------------|--------------------|--------------------------------------------|
| State Machine | ParkingDataManager | Panel present; no behavioral states in XML |
| State Machine | ExitControl        | Behavioral FSM                             |

## Appendix B - Source and Review Boundary

Primary source reviewed: Smart\_Parking\_TTool.xml. The review intentionally follows the terminology and behavior stored in that model. Where the XML contains incomplete text or empty panels, this report identifies the gap instead of silently completing it from external knowledge.